Shamanth M S

shamanthms1@gmail.com • +91 7019378029 • LinkedIn • GitHub • Mysuru, Karnataka

Executive Summary

Third year Machine Learning Engineer focused on building practical AI systems with an emphasis on multimodal learning and reasoning. Proficient in constructing ML models utilizing PyTorch, Scikit-learn, and contemporary MLOps tools including Docker and Kubernetes. Adept at incorporating theoretical knowledge gained from state-of-the-art studies in practice (e.g., document understanding pipelines).

Skills

- **Languages:** Python, JavaScript, SQL
- **Machine Learning:** TensorFlow, Keras, Transformers (HuggingFace), Scikit-Learn, Pandas, NumPy, PyTorch
- **Databases:** PostgreSQL, MySQL
- **MLOps & Deployment:** Docker, Kubernetes, MLflow, Databricks, REST APIs, Model Deployment Pipelines
- **Tools & Concepts:** Git, GitHub

Project Experience

Neuro-Symbolic System-2 Reasoning Architecture for Hallucination Reduction

- Built a hybrid **LLM + ACT-R architecture** to enforce step-by-step reasoning validation and reduce hallucinations in multi-step tasks
- Developed a **Python-based validation layer with HuggingFace Transformers**, improving reasoning consistency on benchmark problems
- Structured **JSON output** using constrained decoding (HuggingFace) to reduce parsing errors by about **30%**
- Designed reasoning workflows using **PyTorch** and data processing pipelines, improving consistency across benchmark tasks **by ~35%**

NyayaFlow — AI-Powered Judicial Case Intelligence System

- Created a judicial case system using AI that runs on **Databricks (Delta Lake, MLflow)** for handling **>10,000** cases in **<2 seconds**
- Trained Random Forest algorithms using Scikit-learn to predict case difficulty and urgency with **~87% accuracy**
- Enhanced precedent search using TF-IDF and embedding similarity to achieve precision of **~0.82**
- Incorporated LLM for automatic brief writing, saving up to **40%** of research time and deployed it using **Kubernetes**

DocuParse AI — Multimodal Document Understanding System

- Developed a multimodal document parser handling more than **1,000 PDF/images**, with **~88% accuracy** in extracting structured data
- Deployed **OCR pipeline** using **Tesseract** with bounding box detection, increasing text extraction accuracy **by ~25%** compared to the baseline
- Utilized **LayoutLM** for layout-based comprehension and worked with a mix of real and synthetic document samples for evaluation
- Experiment tracking with **MLflow**, optimized deployment pipeline and deployment with **Docker**
- Curated dataset using a mix of real and synthetic documents to improve model generalization across varied layouts

Work Experience

AI/ML Intern | Infosys Springboard (Virtual Internship 6.0) — Feb 2026 – Apr 2026

- Automated preprocessing for thousands of records using custom Python scripts, reducing manual data preparation time by **~40%**
- Engineered **14+ new financial risk features** significantly improving model interpretability and establishing baseline accuracy
- Developed a Random Forest model achieving **0.96 ROC-AUC**, documenting the process to establish benchmarks for future iterations
- Collaborated on feature engineering and model evaluation workflows, improving overall pipeline efficiency

Website Developer | Eureka Institute — Nov 2025 – Jan 2026

- Building core frontend modules and UI components for a production-grade platform with responsive, accessible layouts
- Improved page load performance by **~28%** through modular restructuring, optimized state patterns, and reduced render cycles
- Delivering weekly feature updates maintaining a **95% merge acceptance rate** across team reviews

Freelance Web Developer | PROFORMA (German Client) — Oct 2024

- Delivered a full-scale frontend with **13,000+ lines of custom CSS** aligned to enterprise UI standards
- Improved usability by **~45%** through clear navigation flows, optimized interactions, and responsive layouts
- Completed 100% of deliverables within deadlines, achieving strong client satisfaction and **approval in 4 review cycles**

Education

Vidyavardhaka College of Engineering | Mysuru, KA — **Expected August 2027**

Bachelor of Engineering, CSE (AIML) — GPA: 8.22/10

Achievements

- **Runner-Up at TiE U Global Hackathon 2024** and received a VIP invitation to attend the **TiE Global Summit 2025**
- **Won 1st place at Eurekathon 2025** (Coding Track), which led to being offered a **Web Developer role at the Eureka Institute**
- **Finalist at Infosys Global Hackathon 2025** by ranking in the **Top 2 out of 32 teams at the Mysuru DC**, and attended the **Grand Finale in Hyderabad**

Research Papers

- **Wei et al. (2022)** — Chain-of-Thought Prompting Elicits Reasoning in LLMs
- **Yao et al. (2023)** — ReAct: Synergizing Reasoning and Acting in Language Models
- **Lu et al. (2022)** — Unified Multimodal Transformers for Vision-Language Tasks
- **Jaegle et al. (2021)** — Perceiver IO: A General Architecture for Structured Inputs & Outputs
- **Garcez et al. (2019)** — Neuro-Symbolic Learning and Reasoning: A Survey